

IF330 Pemrograman Web.

01 - Intro to Server Side Web Programming

Fenina Adline Twince Tobing

I Putu Ananta Yogiswra

PROGRAM STUDI INFORMATIKA
FAKULTAS TEKNIK DAN INFORMATIKA
UNIVERSITAS MULTIMEDIA NUSANTARA
SEMESTER GANJIL TAHUN AJARAN 2022/2023

Semester Outline

Before Mid:

- 1.Introduction to PHP
- 2.Basic PHP Programming
- 3.PHP Pre-defined Vars & Function, Web Form Processing
- 4.Object Oriented PHP
- 5.File Upload and Processing
- 6.PHP-MySQL Database
- 7.PHP Security

After Mid:

- 1.Introduction to Laravel
- 2.Controller & View
- 3.Model & Migration
- 4.Eloquent Relationship
- 5.Forms & Validation
- 6.Laravel Security
- 7.RESTful API

SCORE

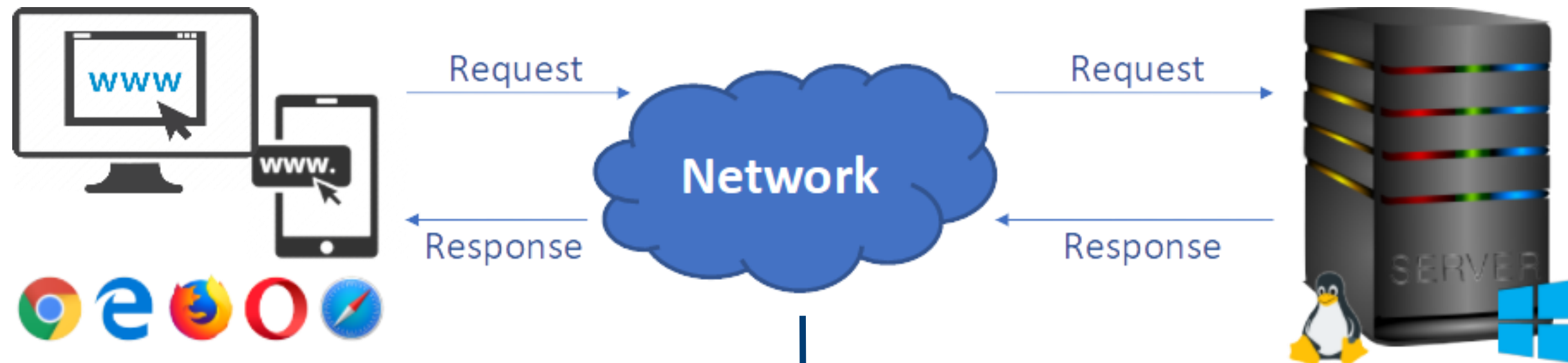
- Mid Exam: 25%
- Final Exam (Final Project): 35%
- Activities: 40%
 - Quiz: 20%
 - Assignment: 10%
 - Proposal Presentation: 10%

TODAY's OUTLINE



- Introduction
- Basic web architecture
- Request-Response Cycle
- Introduction to PHP
- PHP Hello World
- PHP Variables

Basic Web Architecture



Client-Side:

Web Browser:

Chrome / Firefox / Safari / Opera /  / Edge / etc...

Language:

HTML / CSS / JavaScript

Framework & Library:

Bootstrap / Foundation / Angular / ReactJS / etc...

Server-Side:

Web Server:

Apache / Nginx / LiteSpeed / IS / Tomcat

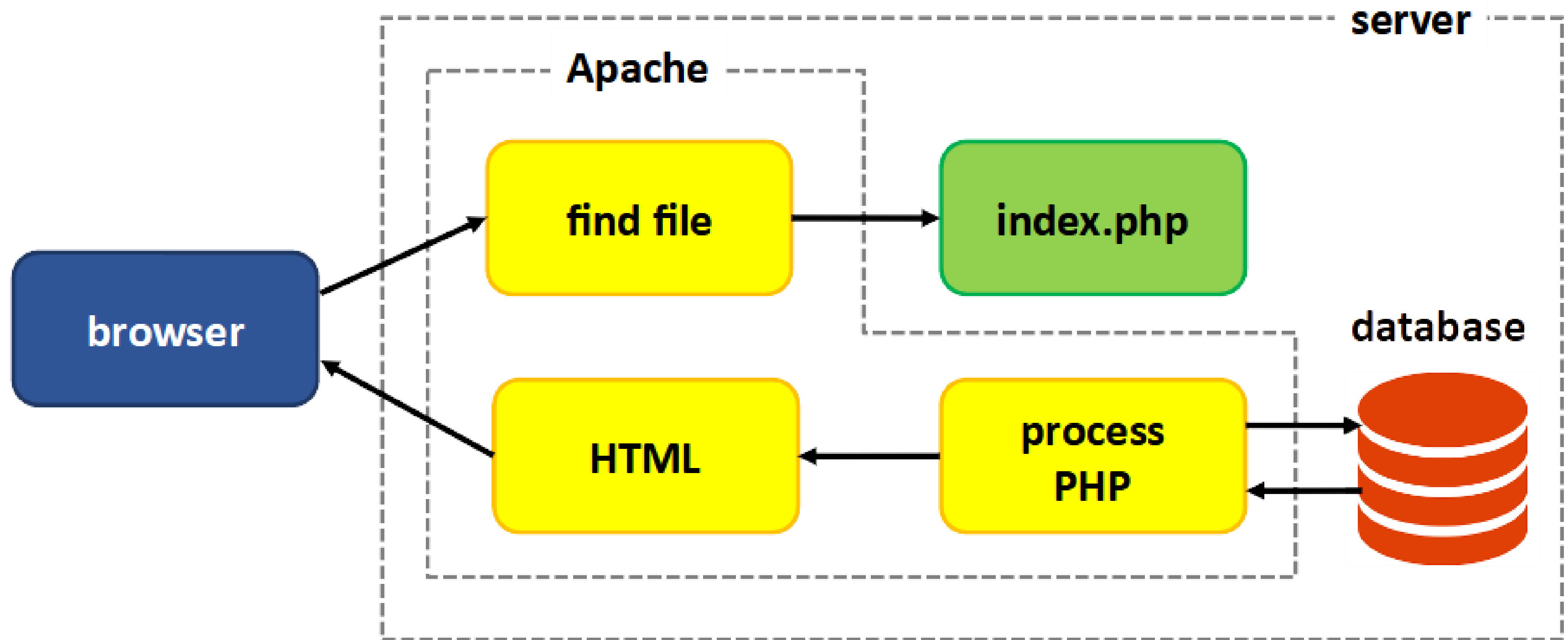
Language:

PHP / APS.Net / Java / Python / JavaScript

Framework:

Laravel / CodeIgniter / Symfony / Node.js / Express / etc...

Request-Response Cycle



Introduction to PHP

PHP is a server-side scripting language designed specifically for the Web. Within an HTML page, you can embed PHP code that will be executed each time the page is visited. Your PHP code is interpreted at the Web server and generates HTML or other output that the visitor will see.



Story Behind PHP



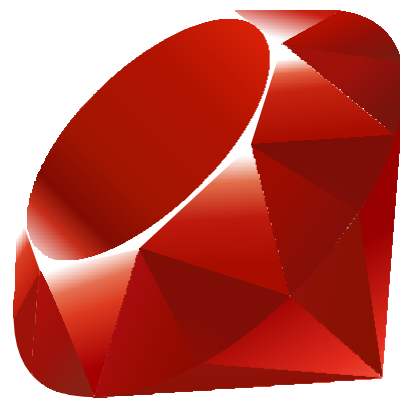
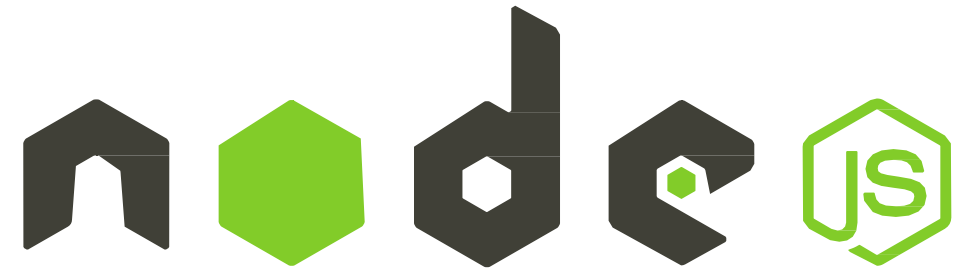
Rasmus Lerdorf

- Rasmus Lerdorf, member of Apache Group
- Develop a tool to track his personal website visitors (1994)
- Publish Personal Home Page Tools (1995)
- Lerdorf was never intentionally develop a new programming language
- But the community develops it organically
- Zeev Suraski and Andi Gutmans formed the base of PHP 3 (1997)
- Change the name to recursive acronym PHP: Hypertext Preprocessor

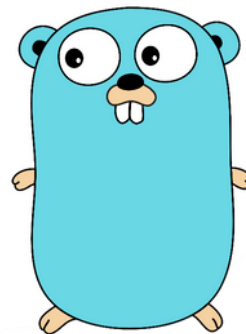
Why PHP?

- **Low barrier to entry.** It's trouble-free to learn with plenty of documentation.
- **Large community.** This means, that if you have problems with PHP, you can immediately find help.
- **Many frameworks and libraries.** Laravel, Phalcon, Symphony, and Zend Framework are the most popular frameworks.
- **Easily embedded into HTML.**
- **Highly scalable.**
- **Operates on various platforms.**

Alternatives to PHP



GOLANG



- **Node.js**
- **Python**
- **Ruby**
- **Golang**
- **Java**
- **C#**

(Good) Topics about PHP

- **Open-source**
- **Easy to use (C-like and Perl-like syntax)**
- **Stable and fast**
- **Multiplatform**
- **Many databases support**
- **Many common built-in libraries**
- **Pre-installed in Linux distributions**

PHP Hello World (1)

- Install Web Server & PHP Interpreter (XAMPP)
- Create "hello.php"
- Put "hello.php" in htdocs folder
- Call "hello.php" from the browser



xampp

...

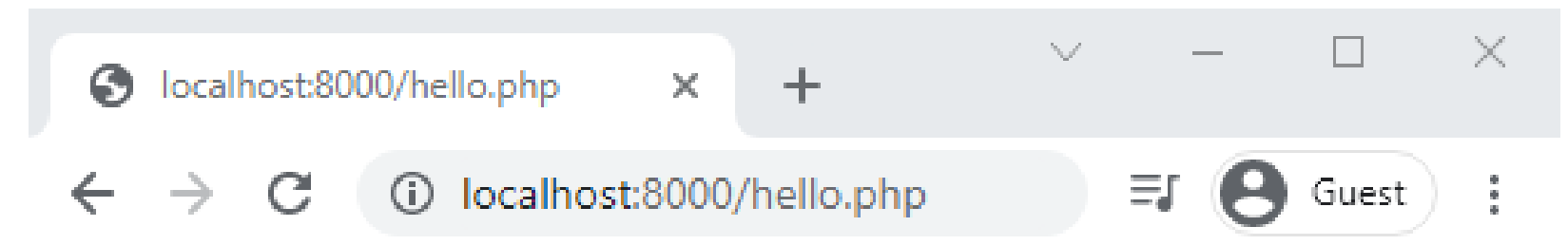
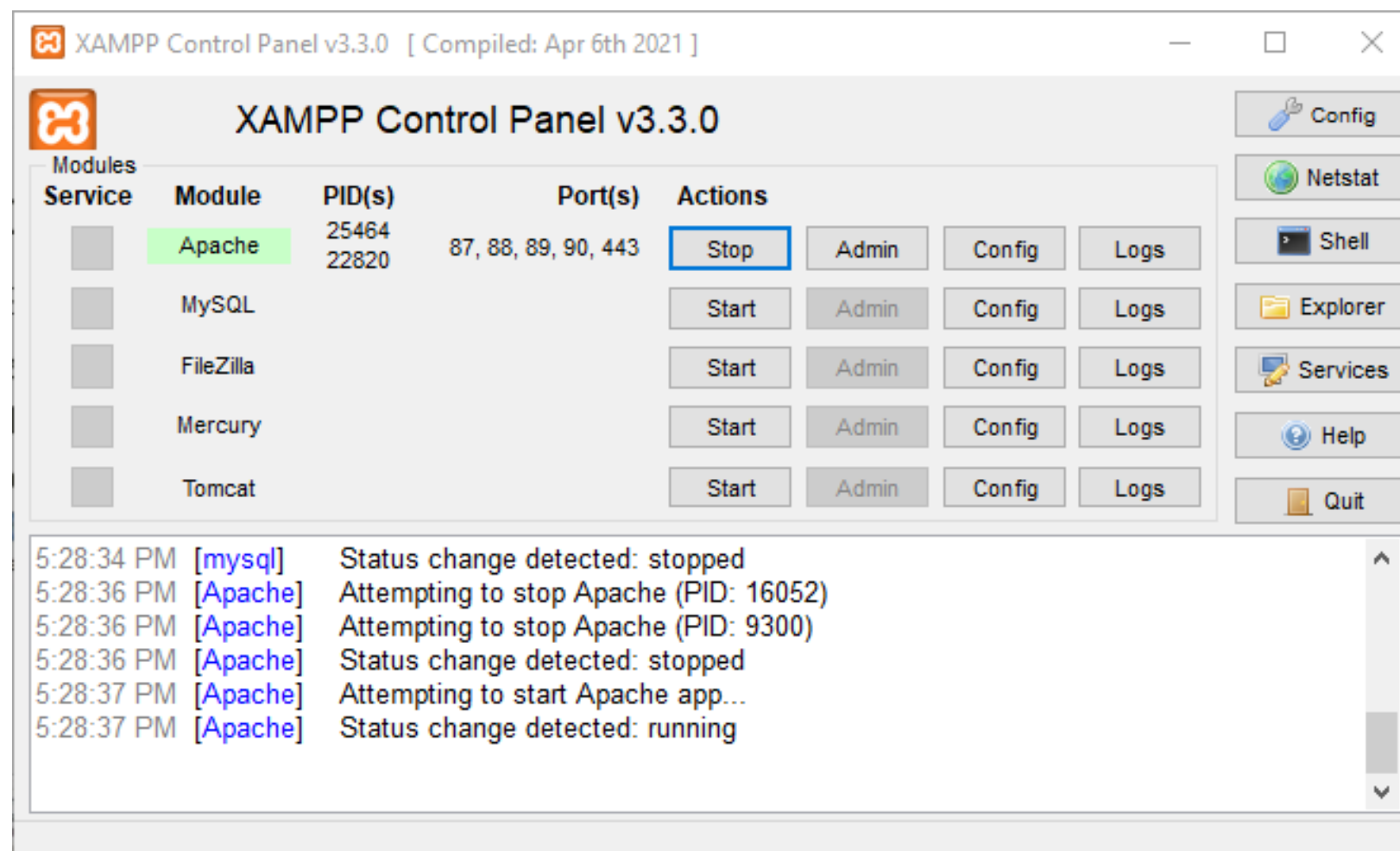


htdocs



hello.php

```
<?php  
echo "<h1>Hello world</h1>";  
?>
```



Hello World

PHP Hello World (2)

```
<!doctype html>
<html>
<head>
  <title>Hello world</title>
</head>
<body>
  <?php echo "Hello world"; ?>
</body>
</html>
```

PHP tag, allow to insert PHP code. Interpretation by PHP module will substitute the code with code output.

Variables

```
$a = 1;  
$b = "2";  
$c = ($num + $text); // integer sum  
$d = $a . $b; // string concatenation  
echo $c; // prints 3  
echo $d; // prints 12
```

- To use or assign variable \$ must be present before the name of the variable
- The assign operator is '='
- There is no need to declare the type of the variable
- The currently stored value produces an implicit type-casting of the variable
- The variable can be used before to be assigned

SUMMARY

After finishing this module, you should be able:

- to understand basic web architecture
- to understand the request-response cycle works
- to understand what is PHP, why we using PHP, alternatives to PHP
- to understand how to create your first PHP code
- to understand how to use variables in PHP

NEXT WEEK'S OUTLINE

- Data types in PHP
- Array
- Functions
- Control structures (if...else, switch..case)
- Loop structures (for, foreach, while, do...while)



next week

REFERENCES

- <https://www.w3schools.com/PHP/default.asp>
- <https://www.php.net/docs.php>
- Luke Welling & Laura Thomson. 2017. PHP and MySQL Web Development (5th ed)
- Robin Nixon. 2021. Learning PHp, MySQL & JavaScript.

Visi

Menjadi Program Studi Strata Satu Informatika **unggulan** yang menghasilkan lulusan **berwawasan internasional** yang **kompeten** di bidang Ilmu Komputer (*Computer Science*), **berjiwa wirausaha** dan **berbudi pekerti luhur**.



Misi

1. Menyelenggarakan pembelajaran dengan teknologi dan kurikulum terbaik serta didukung tenaga pengajar profesional.
2. Melaksanakan kegiatan penelitian di bidang Informatika untuk memajukan ilmu dan teknologi Informatika.
3. Melaksanakan kegiatan pengabdian kepada masyarakat berbasis ilmu dan teknologi Informatika dalam rangka mengamalkan ilmu dan teknologi Informatika.